⛽ Carburetion & Fuel: Reading

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

How a Carburetor Works

An engine needs the right mix of air and fuel to run, about 14.7 parts air to 1 part fuel by weight. The carburetor makes this mix, turns liquid fuel into a fine mist, and controls engine speed.

The key idea is the venturi, a narrow section in the carburetor's throat. As air rushes through the narrow spot, it speeds up and its pressure drops. Normal air pressure on the fuel in the bowl then pushes fuel up through a small jet and into the fast airstream, where it mixes with the air.

Other parts control the mix. The float and needle keep the fuel in the bowl at the correct level. The throttle plate controls how much mixture enters the engine, changing speed. The choke limits air when the engine is cold so the mixture is richer. Float carburetors are used on mowers; diaphragm carburetors, used on chainsaws and trimmers, pump fuel with a flexible diaphragm so they work at any angle.

Fuel quality matters. Most gas contains ethanol, which absorbs water and can leave gum that clogs tiny jets. Use fresh fuel, add stabilizer for storage, and mix two-stroke oil at the correct ratio.

Check your reading

1. Why is the venturi narrow?

2. What does the choke do?

3. Why can ethanol cause problems?